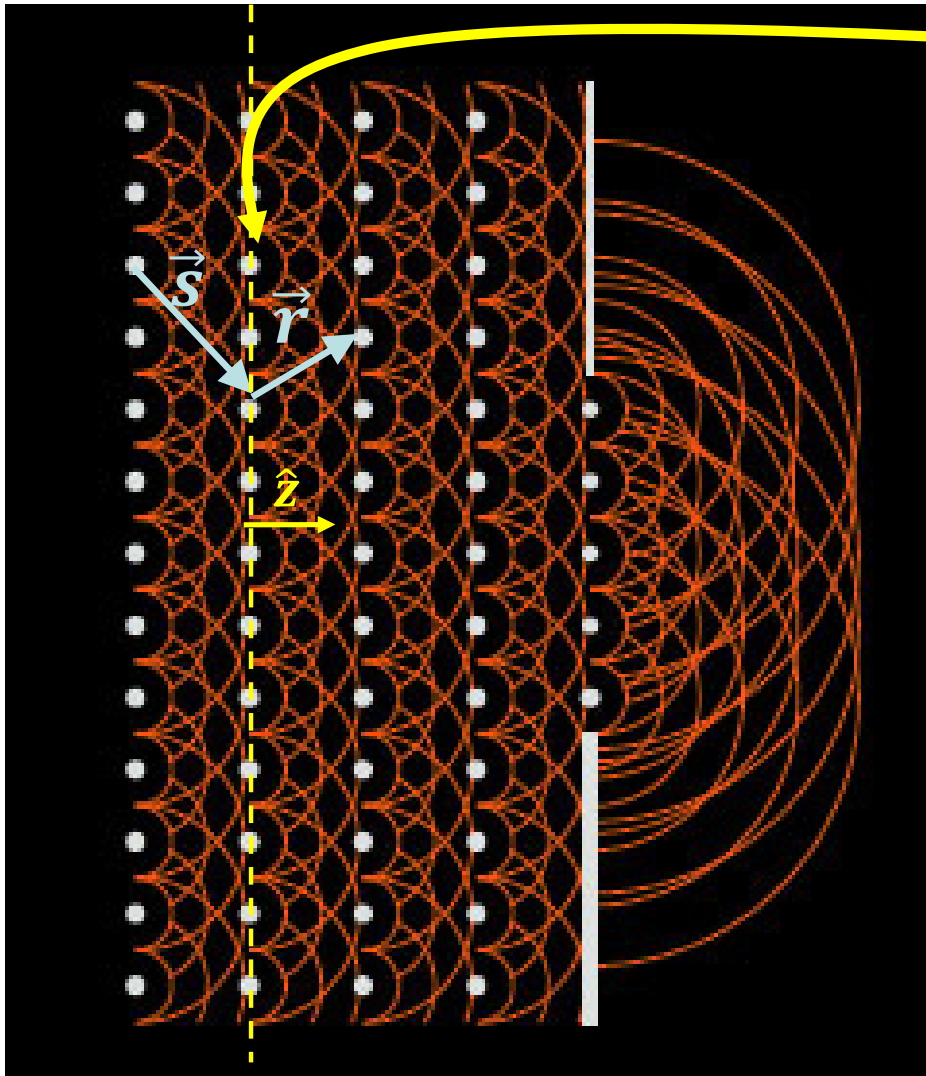


Ph134, Lecture 16

- **Scalar diffraction theory of propagation through an aperture**
 - **Huygens-Fresnel secondary spherical wavelets**
 - **Babinet's principle**
 - **Diffraction of a point source through an aperture, where aperture size is small compared to longitudinal distances from source to aperture and aperture to observation plane.**

Propagation through an aperture: starting from Huygens-Fresnel principle

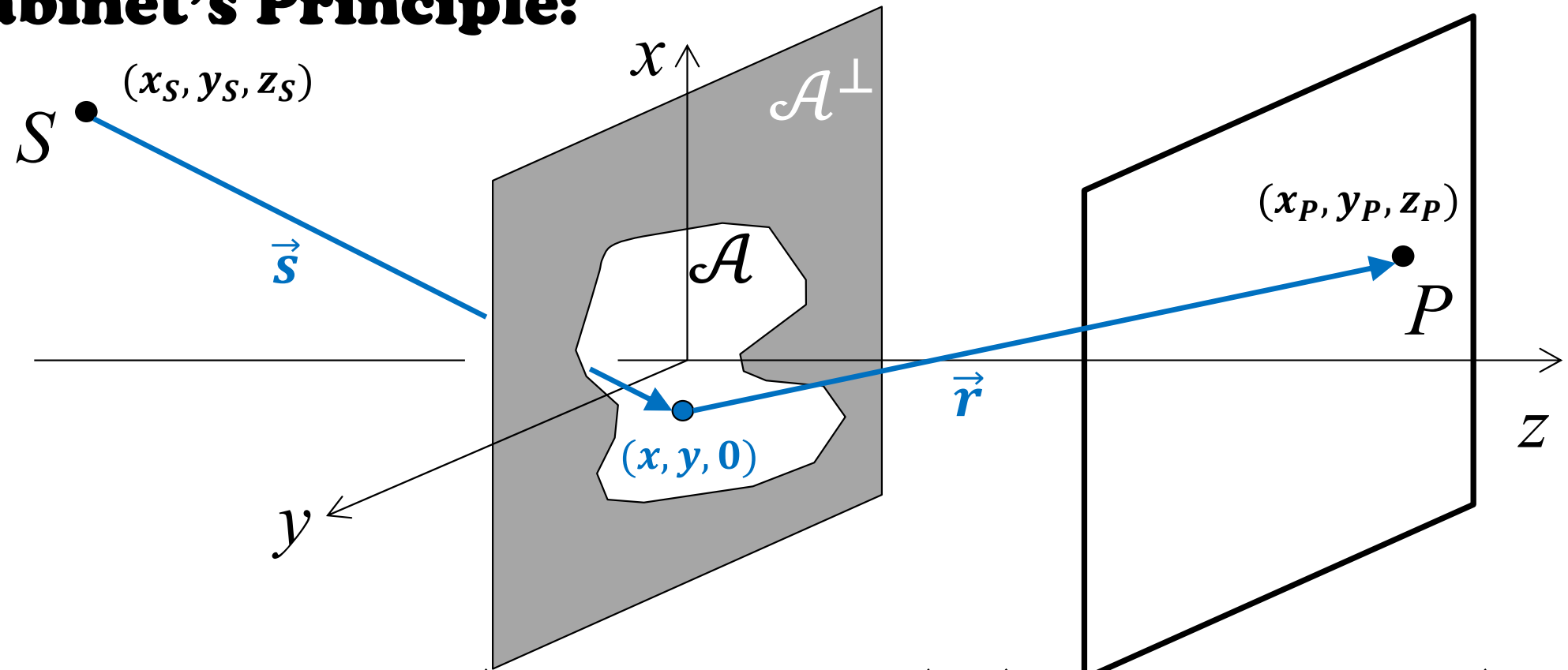
Every unobstructed point of a wavefront, at a given instant in time, serves as a source of spherical secondary wavelets (with the same frequency as that of the primary wave). The amplitude of the optical field at any point beyond is the superposition of all these wavelets (considering their amplitudes and relative phases). -- from Optics, Hecht



Fresnel introduced extra factors required to make the principle quantitatively predict the correct wave propagation in free space: If we are considering one set of spherical wavelets producing a new set at a fixed value of z , and then using those wavelets to find how the field propagates, then the secondary wavelets need this extra factor (the “obliquity factor”):

$$\frac{-i}{\lambda} \frac{[\hat{S} \cdot \hat{z} + \hat{r} \cdot \hat{z}]}{2}$$

Babinet's Principle:



$$u_{\mathcal{A}}(P) = \iint_{\mathcal{A}} dx \, dy \left(\begin{array}{c} \text{original wave} \\ \text{propagated} \\ \text{from } S \text{ to } (x, y, 0) \end{array} \right) \left(\begin{array}{c} \text{secondary wavelet} \\ \text{propagated from} \\ (x, y, 0) \text{ to } P \end{array} \right)$$

$$u_{\mathcal{A}}(P) + u_{\mathcal{A}^\perp}(P) = u_0(P)$$

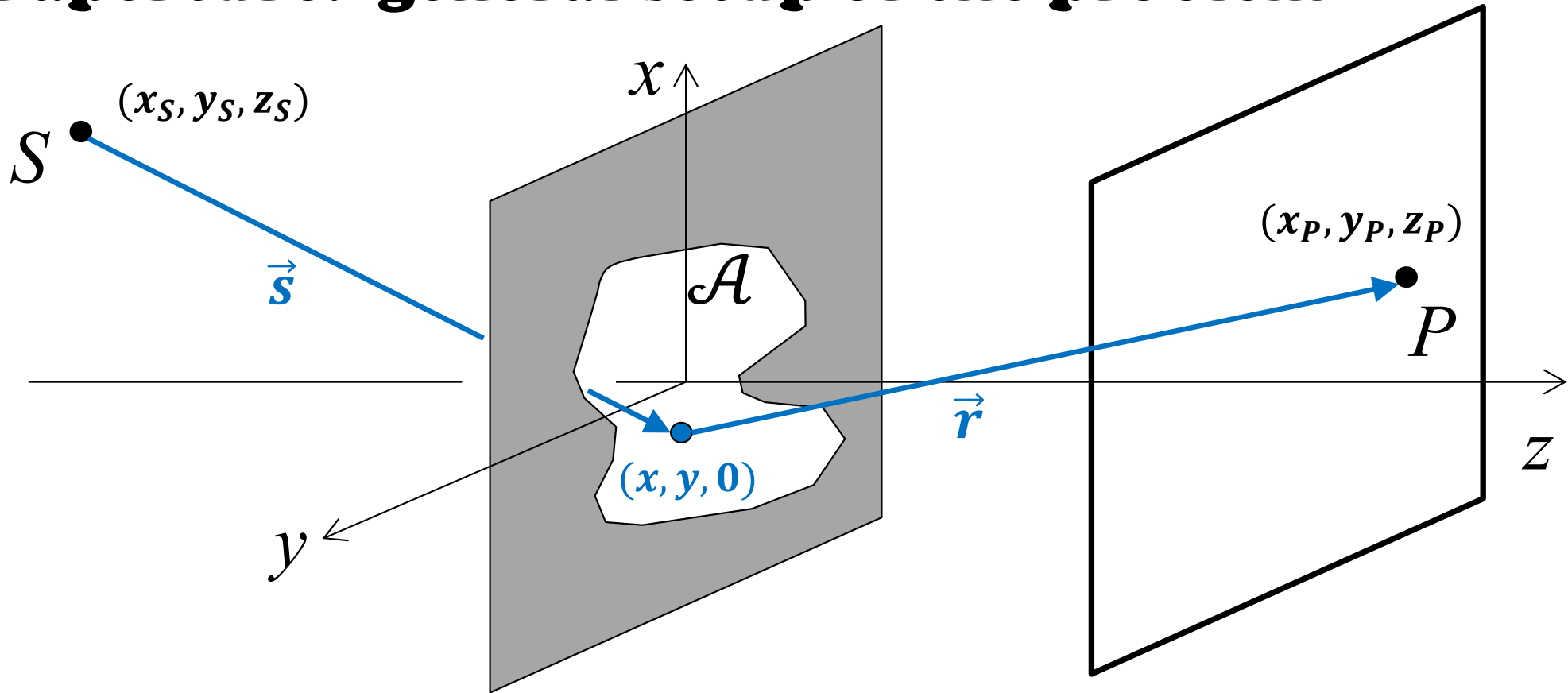
$\uparrow$
Field at P
w/ only $\mathcal{A}$ open

$\uparrow$
Field at P
w/ only $\mathcal{A}$ blocked

$\uparrow$
Field at P
w/ no obstruction

useful when one of the three is zero

Light from point source propagating through an aperture: general setup of the problem

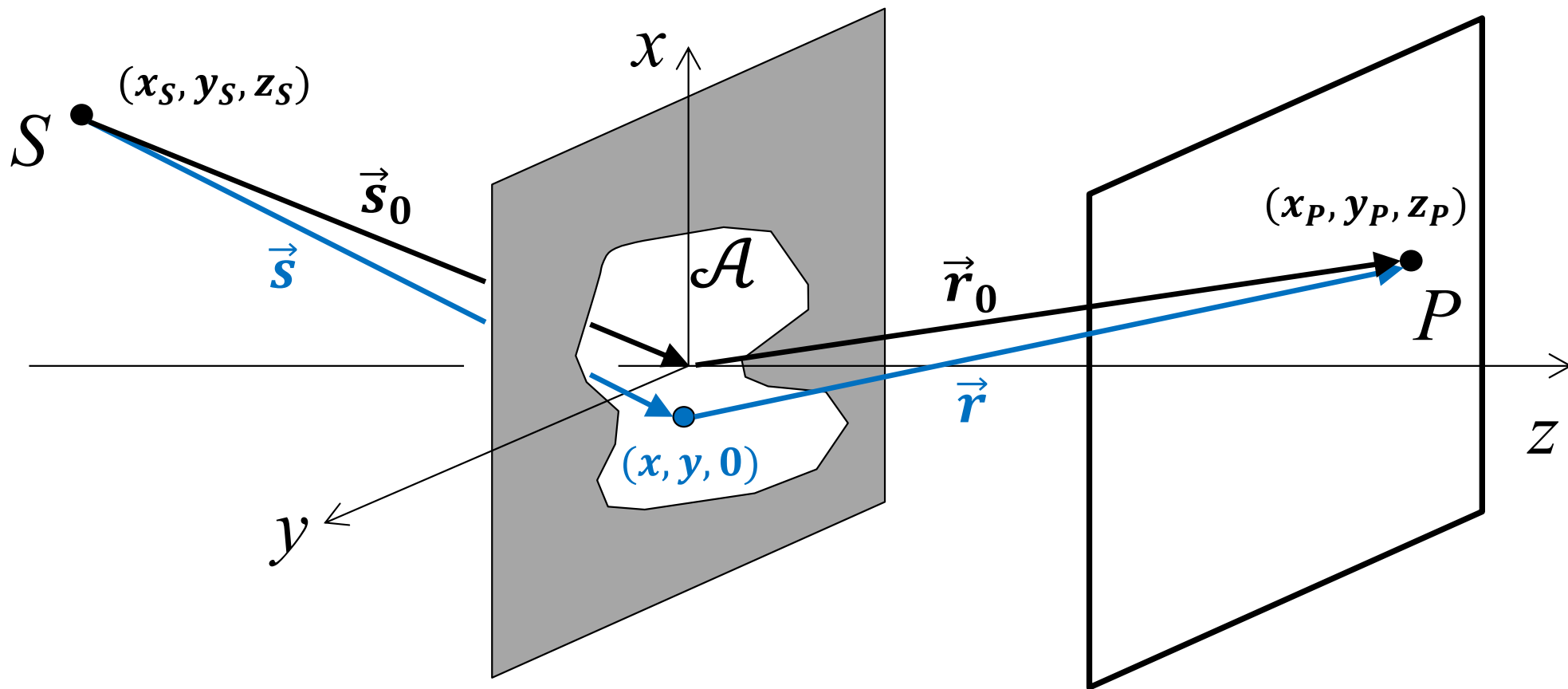


$$E^{(+)} = u(\vec{r})e^{-i\omega t}$$

From Huygens-Fresnel principle:

$$u_{\mathcal{A}}(P) = \iint_{\mathcal{A}} dx \, dy \, A_0^{(+)} \frac{e^{ik|\vec{s}|}}{|\vec{s}|} \frac{e^{ik|\vec{r}|}}{|\vec{r}|} \left(\frac{-i}{\lambda} \frac{[\hat{s} \cdot \hat{z} + \hat{r} \cdot \hat{z}]}{2} \right)$$

Approximations ($|x|, |y| \ll |z_S|, |z_P|$) :

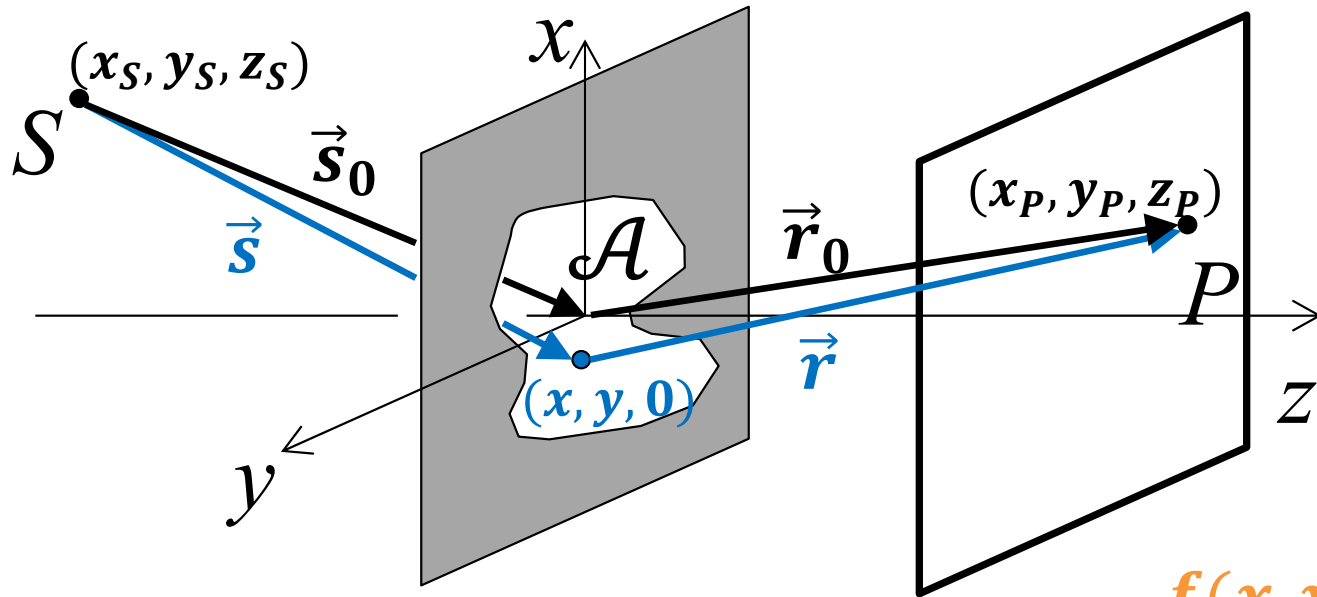


*Keep variation
across aperture here.*

$$u_{\mathcal{A}}(P) = \iint_{\mathcal{A}} dx dy A_0^{(+)} \frac{e^{ik|\vec{s}|}}{|\vec{s}|} \frac{e^{ik|\vec{r}|}}{|\vec{r}|} \left(\frac{-i}{\lambda} \frac{[\hat{S} \cdot \hat{Z} + \hat{r} \cdot \hat{Z}]}{2} \right)$$

Neglect variation across aperture here.

Fraunhofer vs. Fresnel Diffraction:



$$u_{\mathcal{A}}(P) \approx \frac{-iA_0^{(+)} e^{iks_0}}{\lambda s_0} \frac{e^{ikr_0}}{r_0} \frac{[\hat{s}_0 \cdot \hat{z} + \hat{r}_0 \cdot \hat{z}]}{2} \iint_{\mathcal{A}} dx dy e^{ik f(x, y)}$$

$f(x, y) \equiv (s - s_0 + r - r_0)$

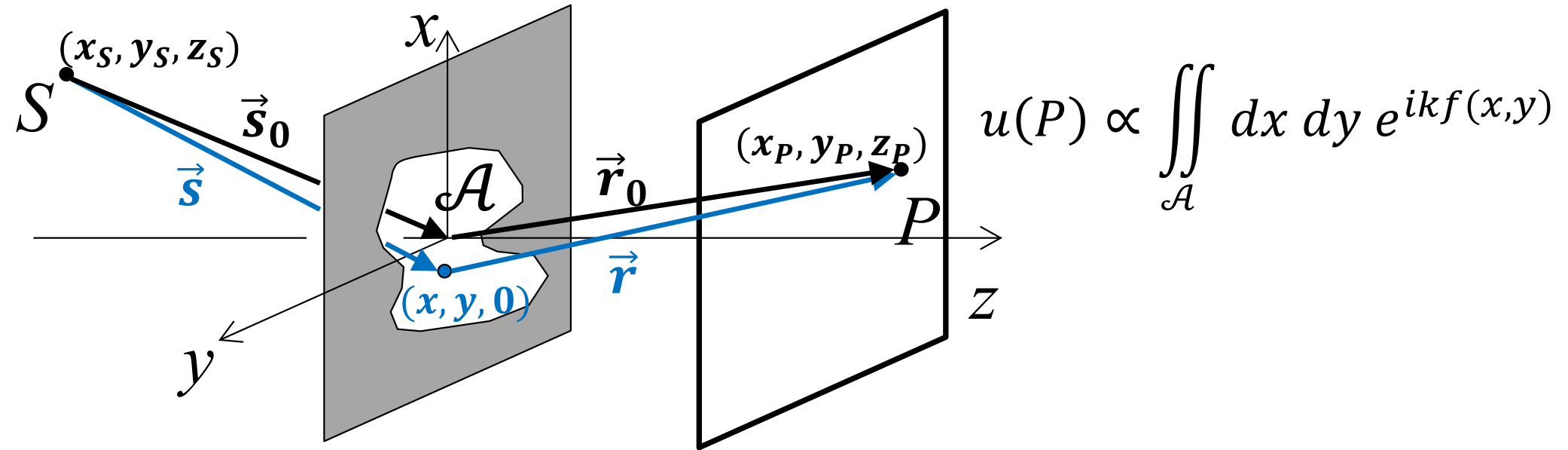
$$f(x, y) \approx -\left(\frac{x_S}{s_0} + \frac{x_P}{r_0}\right)x - \left(\frac{y_S}{s_0} + \frac{y_P}{r_0}\right)y$$

$$+ \frac{1}{2}\left(\frac{1}{s_0} + \frac{1}{r_0}\right)(x^2 + y^2) - \frac{1}{2s_0}\left(\frac{x_S}{s_0}x + \frac{y_S}{s_0}y\right)^2 - \frac{1}{2r_0}\left(\frac{x_P}{r_0}x + \frac{y_P}{r_0}y\right)^2$$

Keep quadratic terms in $x, y \rightarrow$ Fresnel diffraction.

Keep only linear terms in $x, y \rightarrow$ Fraunhofer diffraction.

Physical significance of the terms:



$$u(P) \propto \iint_{\mathcal{A}} dx dy e^{ikf(x,y)}$$

Path length change due to:

$$f(x, y) \approx -\left(\frac{x_S}{s_0} + \frac{x_P}{r_0}\right)x - \left(\frac{y_S}{s_0} + \frac{y_P}{r_0}\right)y$$

Planar wavefronts at an angle to the aperture plane.

$$+ \frac{1}{2} \left(\frac{1}{s_0} + \frac{1}{r_0} \right) (x^2 + y^2)$$

Curved wavefronts at normal incidence.

$$- \frac{1}{2s_0} \left(\frac{x_S}{s_0}x + \frac{y_S}{s_0}y \right)^2 - \frac{1}{2r_0} \left(\frac{x_P}{r_0}x + \frac{y_P}{r_0}y \right)^2$$

Curved wavefronts at an angle to the aperture plane.